No. 4

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# Setup and Configuration of Local Area Network using a Switch and Ethernet cables.

**Aim:**

To setup and configure a local Area Network using a Switch and Ethernet Cables. assign IP addresses and verify communication between computers.

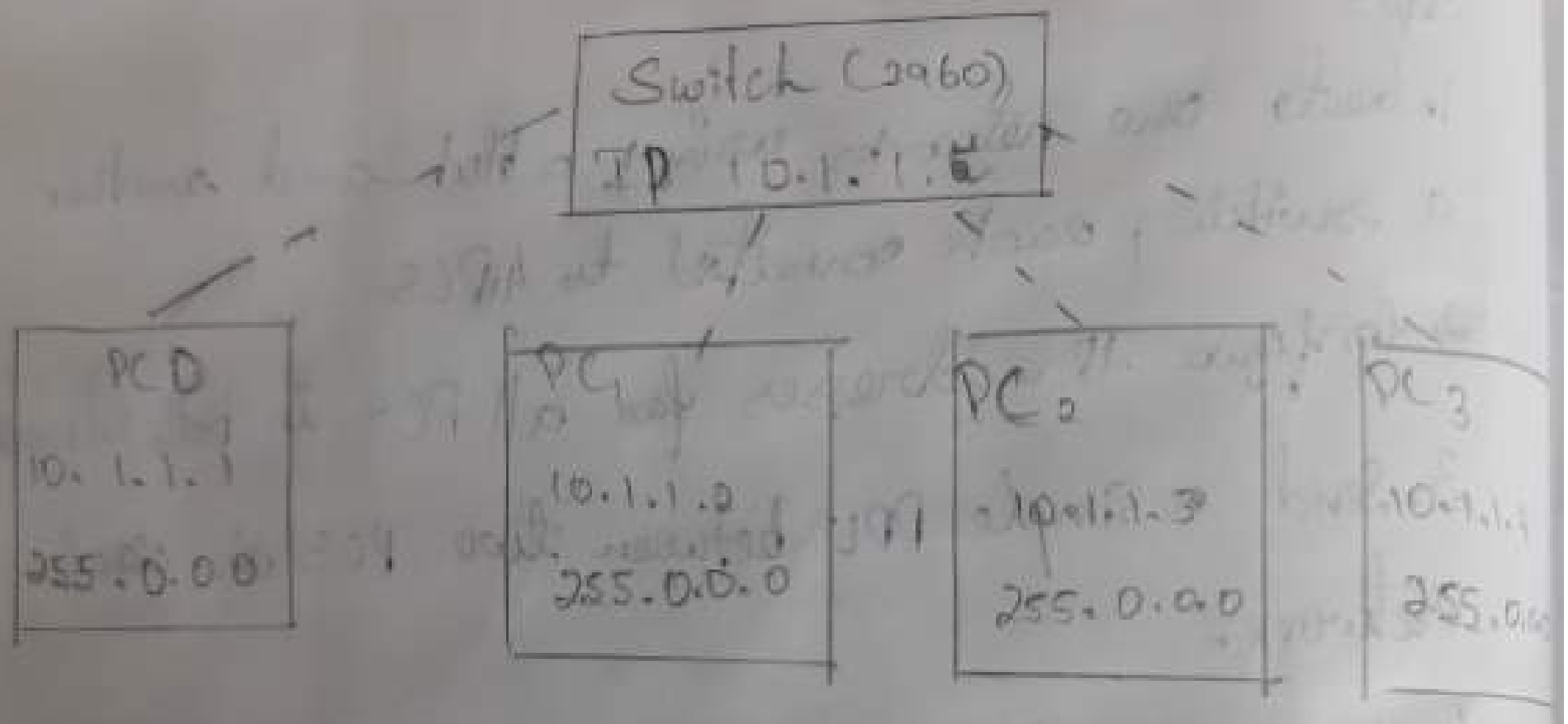
**LAN:**

A Local Area Network connects computers and devices within a small area such as a home, office, or School. It enables users to shares files, Printers, internet access, and other resources, A Switch connects all devices and allows fast and secure communication..

**Step to Set up a LAN:**

1] Plan and Design a appropriate network topology taking into account network requirment and equipment location.

2] Take 4 computers, a switch with 8, 16 or 24 ports which is sufficient for Networks of these sizes and 4 Ethernet Cables.


Switch (2960)
IP 10.1.1.5
PC D
10.1.1.1
255.0.0.0
PC1
10.1.1.2
255.0.0.0
PC0
10.1.1.3
255.0.0.0
PC3

Step 3:- Connect the computer to Network Switch via an Ethernet cables, which is a simple as Plugging. one end of the Ethernet cables into the Computer and other end into the Network Switch.

Step 4:- Assign IP address to your PCs

1. Log on the Client computer as Administrator or as Owner
2. Click Network and Internet connection.
3. Right click Local Area connection / Ethernet → Go to Properties → select Internet Protocol → Click on properties Select use the following ip address option and assign ipaddress.

Similarly assign IP address to all the PCs connected to switch

$PC_1$ - IP address: 10.1.1.1, Subnet mask 255.0.0.0

$PC_2$ IP address - 10.1.1.2, Subnet mask 255.0.0.0

$PC_3$ - IP address - 10.1.1.3, Subnet mask 255.0.0.0

$PC_4$ - IP address - 10.1.1.4, Subnet mask 255.0.0.0

Step -5 :

Configure the switch with the IP address 10.1.1.5.

Step 6 :

Check the connectivity between switch and other machine by using ping command in the Command prompt of the device.

Step 7 :-

Select a folder → go to properties → Click Sharing tab → Share it with everyone. On the Same Lan..

Step 8 :-

Try to access the shared folder from Other Computer of the Networks

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Result :

A LAN was Successfully configured using a Switch and Ethernet cables. IP addresses were assignment correctly, communication was Verified Using the ping command and file sharing was Successfully demonstrated between computerson the LAN.